



Customer Shopping Behavior Analysis

This project analyzes 3,900 customer purchases to uncover spending patterns, preferences, and subscription behavior, guiding strategic business decisions.

Dataset Overview

Dataset Summary

- Rows: 3,900
- Columns: 18
- Missing Data: 37 values in Review Rating

Key Features

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo Code, Previous Purchases, Frequency, Review, Shipping)

Exploratory Data Analysis with Python



We used Python for data preparation and cleaning, including loading, initial exploration, handling missing values, and feature engineering to create new insights.

Data Analysis using SQL: Key Business Insights

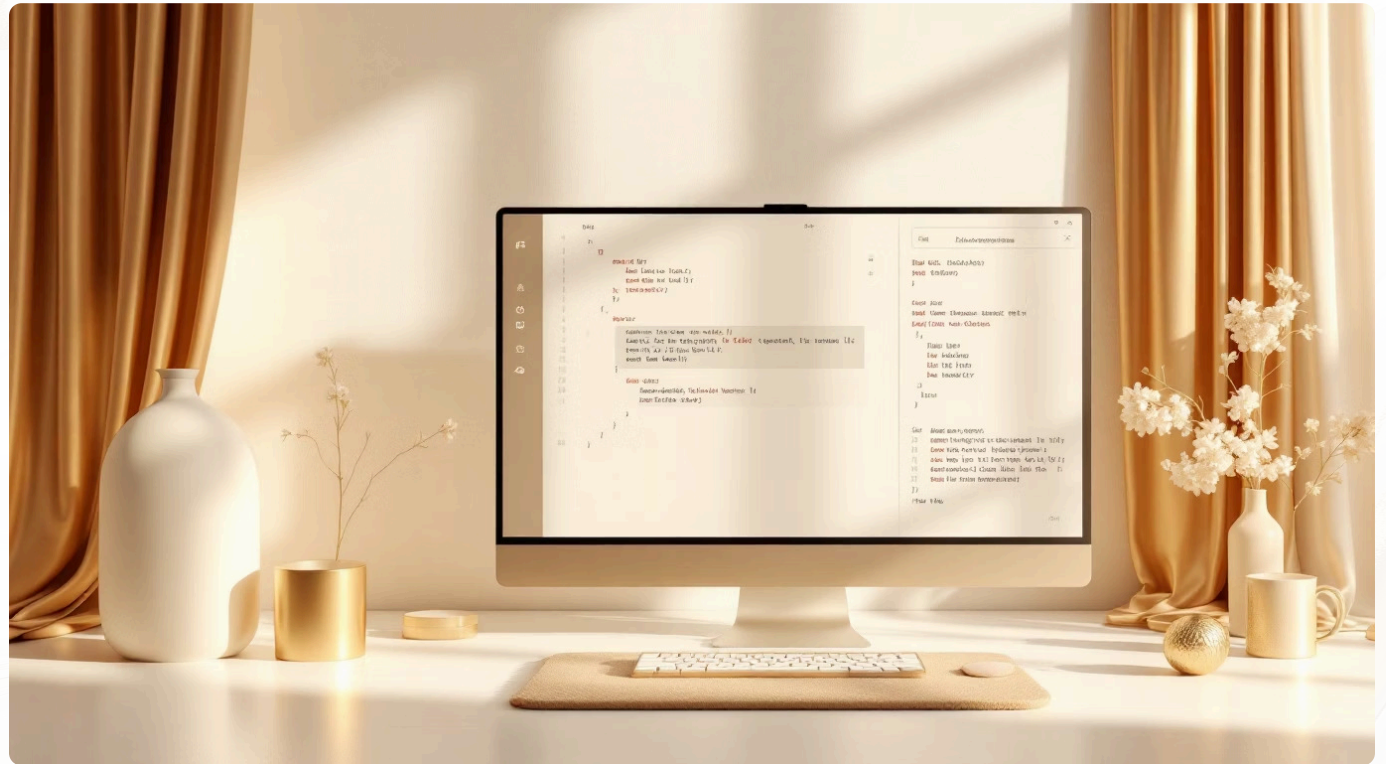
Revenue by Gender

Female: \$75,191

Male: \$157,890

Top 5 Products by Rating

- Gloves: 3.86
- Sandals: 3.84
- Boots: 3.82



Shipping & Subscription Insights

Shipping Type Comparison

Standard: \$58.46 average purchase

Express: \$60.48 average purchase

Subscribers vs. Non-Subscribers

Subscribers: 1,053 customers, \$59.49 avg. spend, \$62,645 total revenue

Non-Subscribers: 2,847 customers, \$59.87 avg. spend, \$170,436 total revenue



Discount & Product Performance



Discount-Dependent Products

- Hat: 50.00% discounted purchases
- Sneakers: 49.66% discounted purchases
- Coat: 49.07% discounted purchases
- Sweater: 48.17% discounted purchases
- Pants: 47.37% discounted purchases

Customer Segmentation



Loyal

3,116 customers



New

83 customers



Returning

701 customers

Customers are classified into segments based on their purchase history, allowing for targeted engagement strategies.

Top Products by Category



Accessories

- Jewelry (171 orders)
- Sunglasses (161 orders)
- Belt (161 orders)



Clothing

- Blouse (171 orders)
- Pants (171 orders)
- Shirt (169 orders)



Footwear

- Sandals (160 orders)
- Shoes (150 orders)
- Sneakers (145 orders)



Outerwear

- Jacket (163 orders)
- Coat (161 orders)

Customer Dashboard Beha

3.90K

total_Customer

\$59.76

average_purchase_amount

r by subscription_status



No of customer by category



e_group



Sales by age



Power BI Dashboard: Visualizing Insights

An interactive Power BI dashboard provides a visual summary of key metrics, customer demographics, and revenue distribution by category and age group.

Strategic Business Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber base.

Customer Loyalty Programs

Reward repeat buyers to foster loyalty and retention.

Review Discount Policy

Optimize discount strategies to balance sales and profit margins.

Targeted Marketing

Focus campaigns on high-revenue age groups and express shipping users.